14 March 2026

Exercise 2: Aggregate Function & Grouping

1. COUNT( ) Function

Write a SQL query to find the total number of employees in the company.

```
SELECT COUNT(id)
FROM employees;
```

id
10

2. SUM( ) Function

Write a SQL query to find total salary paid to all employees in the IT department.

```
SELECT SUM(salary) AS IT_total_salary
FROM employees
WHERE department = 'IT';
```

IT_total_salary
220 000

3. AVG( ) Function

Write a SQL query to calculate the average salary paid to employees in the HR department

```
SELECT AVG(salary) AS HR_Average_salary
FROM employees
WHERE department = 'HR';
```

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HR_Average_Salary
49500

4. Min() and Max() Functions

Write a SQL query to find the highest and lowest salary in the company.

```
SELECT MIN(salary) AS Lowest_salary,
       MAX(salary) AS Highest_salary
FROM employees;
```

Lowest_salary	Highest_salary
48000	62000

5. GROUP BY Statement

Write a SQL query to group employees by department and display the total salary paid in each department.

```
SELECT department,
       SUM(salary) AS Total_salary
FROM employees
GROUP BY department;
```

department	Total_salary
IT	220 000
HR	99000
Finance	119000
Marketing	105000

6. GROUP BY and COUNT()

Write a SQL query to count how many employees [illegible] each city

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```
SELECT city,
       COUNT(id)
FROM employees
GROUP BY city;
```

city	count(id)
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

```
SELECT department,
       AVG(salary) AS Average_salary
FROM employees
GROUPBY department
ORDER BY Average_salary DESC;
```

department	Average_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

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8. HAVING Clause

Write a SQL to find departments where the total salary paid exceeds 100,000. (USE GROUP BY and HAVING)

```
SELECT department,
       SUM(salary) AS Total_salary
FROM employees
GROUP BY department
HAVING Total_salary > 100 000;
```

department	Total_salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY

Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
SELECT city,
       COUNT(id) AS Employee_count
FROM employees
GROUP BY city
HAVING Employee_count > 1
ORDER BY Employee_count DESC;
```

City	Employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

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10. Combining Aggregate Functions

Write a SQL query to find the department with the highest average salary.

```
SELECT department,
       AVG(salary) AS Highest_Average_Salary
FROM employees
GROUP BY department
ORDER BY Highest_Average_salary DESC
LIMIT 1;
```

department	Highest_Average_salary
Finance	59500

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